IGCSE 0607 Practice Questions — E1.8 — Combined Questions & Markscheme

Standard form

Total Questions: 15 — **Total Marks:** 81 Marks

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  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 **(a)** Write 450000 in standard form.

..... **[1]**

(b) Write 3.08×10^{-4} as an ordinary number.

..... **[1]**

(c) Work out.

$(2 \times 10^3) \times (4 \times 10^5)(0)$

Give your answer in standard form.

..... **[1]**

2 **(a)** Write 452000 in standard form.

..... **[1]**

(b) Write 6.07×10^{-3} as an ordinary number.

..... **[1]**

(c) Work out $(4 \times 10^4) \times (3 \times 10^5)$.
Give your answer in standard form.

..... **[2]**

3 **(a)** Write 0.00714 in standard form.

..... **[1]**

(b) Write 5.02×10^4 as an ordinary number.

..... **[1]**

(c) Work out the value of

$$(3 \times 10^5) \times (6 \times 10^4)$$

Give your answer in standard form.

..... **[2]**

4 Consider the following numerical values.

(a) Convert the following numbers as instructed.

(i) Write 0.00034 in standard form.

..... **[1]**

(ii) Write 5.2×10^4 as an ordinary number.

..... **[1]**

(b) Work out, giving your answer in standard form:

$(2 \times 10^3) \times (4 \times 10^5)$

..... **[2]**

5 **(a)** Write 0.000452 in standard form.

..... **[1]**

(b) Write 3.71×10^5 as an ordinary number.

..... **[1]**

(c) Calculate, giving your answer in standard form.

$(2 \times 10^4) \times (4 \times 10^3)$

..... **[2]**

6 The mass of the Earth is 5.97×10^{24} kg.
The mass of the Moon is 7.35×10^{22} kg.

(a) Calculate the difference between the mass of the Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) The mass of the Sun is 1.99×10^{30} kg.
Find the value of k when

mass of the Sun = k ×mass of the Earth
Give your answer to 3 significant figures.

$k =$ ..... **[3]**

7 **(a)** Work out, giving your answer in standard form.

$$\frac{3.6 \times 10^{-4}}{8 \times 10^{-7}}$$

..... **[3]**

(b) Work out, giving your answer in standard form.

$$(5 \times 10^{8}) + (7 \times 10^{7})$$

..... **[2]**

(c) Given that $x = 2 \times 10^{m}$ and $y = 8 \times 10^{n}$, where m and n are integers.
Find an expression for x^2y in standard form, in terms of m and n.

..... **[2]**

8 **(a)** Work out the exact value of

$$\frac{5.4 \times 10^5 + 6 \times 10^4}{2 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

(b) Evaluate $\sqrt{2.5 \times 10^9}$.
Give your answer in standard form.

..... **[2]**

(c) Given that $p = 2 \times 10^x$ and $q = 8 \times 10^y$, find an expression for $p^2 \div q$.
Give your answer in standard form in terms of x and y.

..... **[3]**

9 $p = 5 \times 10^m$ and $q = 8 \times 10^n$, where m and n are integers.

(a) Find an expression, in terms of m and n, for $p \times q$.
Give your answer in standard form.

..... **[2]**

(b) Find an expression, in terms of m and n, for $\frac{p^2}{q}$.
Give your answer in standard form.

..... **[3]**

(c) Given that $p + q = 5.08 \times 10^m$, work out the exact value of $m - n$.

$m - n =$ ..... **[3]**

10 The distance from the Earth to the Sun is 1.5×10^8 km.

(a) Light travels at a speed of 3.0×10^5 km/s.
Calculate the time taken for light to travel from the Sun to the Earth.
Give your answer in seconds.

..... seconds **[2]**

(b) A spacecraft travels at a constant speed of 2.5×10^4 km/h.
Work out the time it takes the spacecraft to travel from the Earth to the Sun.
Give your answer in hours, in standard form.

..... hours **[2]**

11 The mass of the Sun is 1.99×10^{30} kg.
The mass of Jupiter is 1.90×10^{27} kg.

(a) Calculate how many times greater the mass of the Sun is than the mass of Jupiter.
Give your answer in standard form correct to 3 significant figures.

..... **[3]**

(b) A comet has a mass of 4.5×10^{13} kg.
Calculate the mass of the comet in grams.
Give your answer in standard form.

..... g **[2]**

12 **(a)** Calculate, giving your answer in standard form.

$3.4 \times 10^7 + 2.1 \times 10^6$

..... **[2]**

(b) A computer can do a calculation in 4×10^{-9} seconds.
Work out how many of these calculations the computer can do in 5 minutes.
Give your answer in standard form.

..... **[3]**

13 The mass of the Earth is approximately 5.97×10^{24} kg.
The mass of the Moon is approximately 7.35×10^{22} kg.

(a) Calculate the total mass of the Earth and the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) Calculate how many times greater the mass of the Earth is than the mass of the Moon.
Give your answer correct to the nearest integer.

..... **[2]**

(c) The mass of a planet is $p \times 10^{25}$ kg.
The mass of this planet is 300 times the mass of the Earth.
Work out the value of p, giving your answer correct to 3 significant figures.

$p =$ ..... **[2]**

14 **(a)** The mass of Earth is 5.97×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg.
Calculate the difference between the mass of Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) The distance from the Sun to Earth is 1.50×10^{8} km.
The speed of light is 3.00×10^{5} km/s.

Calculate the time it takes for light to travel from the Sun to Earth.
Give your answer in standard form.

..... s **[2]**

(c) The Earth is modelled as a sphere with radius $r = 6.37 \times 10^{6}$ m.
Calculate the surface area of the Earth.
Give your answer in standard form, correct to 3 significant figures.
The surface area, A, of a sphere with radius r is $A = 4\pi r^2$.

..... m^2 **[3]**

15 A large cylindrical water tank has a base radius of 4.5×10^2 cm and a height of 1.2×10^3 cm.

(a) Calculate the volume of the water tank in cm^3.
Give your answer in standard form correct to 4 significant figures.

..... cm^3 **[3]**

(b) The empty tank is being filled with water at a constant rate of 1.5×10^{-4} m^3 per second.
Note that $1\ m^3 = 1 \times 10^6\ cm^3$.

Calculate the time taken to fill the tank completely.
Give your answer in hours, written in standard form correct to 3 significant figures.

..... hours **[4]**

1 **(a)** 4.5×10^5 **[B1]**

(b) 0.000308 **[B1]**

(c) 8×10^8 **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 4.52×10^5 **[B1]**

(b) 0.00607 **[B1]**

(c) 1.2×10^{10} **[B2]**

If 0 scored, M1 for 12×10^9 **[M1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** 7.14×10^{-3} **[B1]**

(b) 50200 **[B1]**

(c) 1.8×10^{10} **[B2]**

M1 for 18×10^9 or 18000000000 seen

[Back to Question] *[Total Marks: 4]*

4 **(a)** **(i)** 3.4×10^{-4} **[B1]**

(ii) 52000 **[B1]**

(b) 8×10^8 **[B2]**

If B0, M1 for $2 \times 4 = 8$ or $10^3 \times 10^5 = 10^8$ seen.

[Back to Question] *[Total Marks: 4]*

5 **(a)** 4.52×10^{-4} **[B1]**

(b) 371000 **[B1]**

(c) 8×10^7 **[B2]**

If 0 scored, allow [M1] for 80000000 or 8×10^k where $k \neq 7$

[Back to Question] *[Total Marks: 4]*

6 **(a)** $597 \times 10^{22} - 7.35 \times 10^{22}$ or $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ **[M1]**

5.8965×10^{24} **[A1]**

(b) Sets up the correct division: $\dfrac{1.99 \times 10^{30}}{5.97 \times 10^{24}}$ **[M1]**

Evaluates correctly, e.g., $0.3333... \times 10^6$ **[M1]**

3.33×10^5 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** 4.5×10^2 **[B3]**

If 0 scored, allow [M1] for $\frac{3.6}{8} \times 10^3$ or 0.45×10^3

Allow a further [M1] for correctly converting to standard form, or for seeing 450

(b) 5.7×10^8 **[B2]**

If 0 scored, allow [M1] for $(50 \times 10^7) + (7 \times 10^7)$ or 570000000 or digits 57

(c) $3.2 \times 10^{2m+n+1}$ **[B2]**

If 0 scored, allow [M1] for $(4 \times 10^{2m}) \times (8 \times 10^n)$ or $32 \times 10^{2m+n}$

[Back to Question] *[Total Marks: 7]*

8 **(a)** 3×10^8 **[B3]**

If 0 scored, M1 for $5.4 \times 10^5 + 0.6 \times 10^5$ or $54 \times 10^4 + 6 \times 10^4$ or 6.0×10^5 seen **[M1]**

M1 for $\frac{\text{their } 6.0 \times 10^5}{2 \times 10^{-3}}$ **[M1]**

(b) 5×10^4 **[B2]**

If 0 scored, M1 for $\sqrt{25 \times 10^8}$ or 50000 seen **[M1]**

(c) $5 \times 10^{2x-y-1}$ **[B3]**

If 0 scored, M1 for $p^2 = 4 \times 10^{2x}$ **[M1]**

M1 for $\frac{4 \times 10^{2x}}{8 \times 10^y}$ or $0.5 \times 10^{2x-y}$ **[M1]**

[Back to Question] *[Total Marks: 8]*

9 **(a)** $4 \times 10^{m+n+1}$ or $4.0 \times 10^{m+n+1}$ **[B2]**

M1 for $40 \times 10^{m+n}$ seen

(b) $3.125 \times 10^{2m-n}$ **[B3]**

M1 for $(5 \times 10^m)^2$ or 25×10^{2m} seen

M1 for $\frac{25}{8} \times 10^{2m-n}$

(c) 2 **[B3]**

M1 for $8 \times 10^n = 5.08 \times 10^m - 5 \times 10^m$

M1 for $8 \times 10^n = 0.08 \times 10^m$ or $8 \times 10^n = 8 \times 10^{m-2}$

[Back to Question] *[Total Marks: 8]*

10 **(a)** 500 **[B2]**

M1 for $1.5 \times 10^8 \div 3.0 \times 10^5$

(b) 6.0×10^3 or 6×10^3 **[B2]**

M1 for $1.5 \times 10^8 \div 2.5 \times 10^4$ or 6000 seen

[Back to Question] *[Total Marks: 4]*

11 **(a)** $\dfrac{1.99 \times 10^{30}}{1.90 \times 10^{27}}$ **[M1]**

$1047.36\ldots$ or $1.047\cdots \times 10^3$ **[A1]**

1.05×10^3 **[A1]**

(b) $(4.5 \times 10^{13}) \times 1000$ or $(4.5 \times 10^{13}) \times 10^3$ **[M1]**

4.5×10^{16} **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** 3.61×10^7 **[B2]**

If B0, M1 for $3.4 \times 10^7 + 0.21 \times 10^7$ or $34 \times 10^6 + 2.1 \times 10^6$ or 36100000 **[M1]**

(b) Converts minutes to seconds: $5 \times 60 = 300$ **[M1]**

Sets up division: $\dfrac{300}{4 \times 10^{-9}}$ **[M1]**

7.5×10^{10} **[A1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** 6.04×10^{24} **[B2]**

If 0 scored, allow [M1] for 6.0435×10^{24} or 604.35×10^{22} or digits 604

(b) 81 **[B2]**

If 0 scored, allow [M1] for $\dfrac{5.97 \times 10^{24}}{7.35 \times 10^{22}}$ or 81.2...

(c) 1.79 **[B2]**

If 0 scored, allow [M1] for $300 \times 5.97 \times 10^{24}$ or 1.791×10^{27} or digits 179

[Back to Question] *[Total Marks: 6]*

14 **(a)** 5.8965×10^{24} **[B2]**

If 0 scored, M1 for $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ or $597 \times 10^{22} - 7.35 \times 10^{22}$ seen **[M1]**

(b) 5×10^2 or 5.00×10^2 **[B2]**

If 0 scored, M1 for $\dfrac{1.50 \times 10^8}{3.00 \times 10^5}$ or 500 **[M1]**

(c) 5.10×10^{14} **[B3]**

If 0 scored, M2 for $4 \times \pi \times (6.37 \times 10^6)^2$ **[M2]**

If 0 or M2 not scored, M1 for $(6.37 \times 10^6)^2$ or $4.057... \times 10^{13}$ **[M1]**

[Back to Question] *[Total Marks: 7]*

15 (a) $\pi \times (4.5 \times 10^2)^2 \times (1.2 \times 10^3)$ **[M1]**

$7.63407 \cdots \times 10^8$ **[A1]**

7.634×10^8 **[A1]**

(b) Volume in $m^3 = \dfrac{7.634 \times 10^8}{10^6} = 763.4\ m^3$ (or equivalent rate conversion $1.5 \times 10^2\ cm^3/\text{s}$) **[M1]**

Time in seconds $= \dfrac{763.4}{1.5 \times 10^{-4}} = 5.089 \cdots \times 10^6$ **[M1]**

Time in hours $= \dfrac{5.089 \cdots \times 10^6}{3600} = 1413.7 \ldots$ **[M1]**

1.41×10^3 **[A1]**

[Back to Question] *[Total Marks: 7]*